

Hitchhiking Hands: Remote Interaction by Switching Multiple Hand Avatars with Gaze

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Figure 1: Applications of Hitchhiking Hands. Left: Placing a virtual object (a teddy bear) on a faraway shelf. Center: Switching smoothly between and operate XR apps. Right: Operating a remote robotic arm with an AR hand by looking at it (white arrow).

ABSTRACT

Virtual reality (VR) and augmented reality (AR) with eye tracking and hand tracking are widely used in entertainment, gaming, design, and training. However, most VR and AR interaction methods are limited in their interactable range and do not fully support the direct manipulation of VR and AR objects. This paper proposes a novel technique called Hitchhiking Hands, which allows the user to switch multiple hand avatars by staring them, and enables natural and direct interaction with VR/AR objects ranging from nearby objects to remote ones.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality; Virtual reality.

KEYWORDS

virtual hands, virtual reality, augmented reality, 3DUI

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1 INTRODUCTION

Eye tracking and hand tracking in virtual reality (VR) and augmented reality (AR) has been adopted as interaction methods for many applications. Advanced hand tracking enables natural and effective “direct hand manipulation,” with which users can reach out, grasp an object directly with their hand, and move it around in the virtual environment (VE) with natural movements. However, this intuitive approach has many limitations. First, the range of motion of the physical hand is limited to a small space around the user’s body. Thus, it can only grasp nearby objects, and the range of movement of objects is also limited. The problem can be solved by moving the user to a position where they can grasp an object at a distance, but the movement should be optional.

As a prior technique for manipulating remote objects without moving, the go-go interaction technique [Poupyrev et al. 1996] has been proposed. It enables seamless and direct manipulation of both nearby and distant objects using a metaphor that interactively grows the user’s arm and nonlinear mapping. However, it suffers from reduced selection precision as the distance to the target increases. HOMER [Bowman and Hodges 1997] addresses this problem but has difficulty manipulating distant objects precisely and moving near objects far away. Scaled-world grab [Mine et al. 1997] solved this issue by scaling the world down automatically around the user’s head every time he/she grabs an object and scales back up when he/she releases it. However, this technique modifies the scale of the virtual objects and cannot be used in AR situations. Recent techniques that combine gaze and pinch [Pfeuffer et al. 2017; Yu et al. 2021] are promising; it has disadvantages because it is not direct hand manipulation, making it unsuitable for interactions with tools, and it does not address operations in the absence of a gaze target like sculpting. In addition, there is a problem of hand

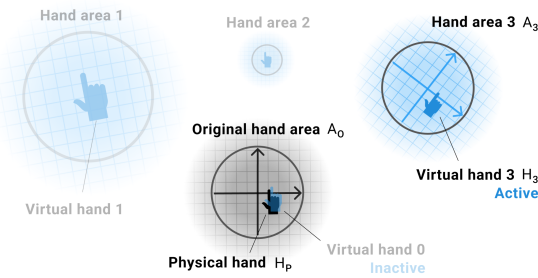


Figure 2: Schematic figure of Hitchhiking Hands.

fatigue in VR/AR direct hand manipulation, partly due to the unrestricted interaction range; large manipulations require users to move the hand large too. In this study, we propose Hitchhiking Hands, which uses gaze and hands to solve these issues.

2 SYSTEM

Our proposed system, *Hitchhiking Hands*, allows users to freely possess several virtual hands placed in the environment using their gaze. Several “hand areas” are placed at regions of interest in the environment, and each hand area has its own virtual hand (Figure 2). When activated, each virtual hand moves in a position and posture that maintains its relative position within the hand area. The original hand area is set within the operating range of the user’s arm for comfort. Other hand areas face the user’s direction for natural movement. The flow of manipulation is as follows.

- (1) An active virtual hand is shown at the physical hand position.
- (2) By staring at one of the deactivated virtual hands, the user can activate that hand and deactivate the previous hand.
- (3) The user can move the active virtual hand with scaled physical hand motion according to the size of the hand area.
- (4) By grabbing an object with the active hand and staring at another virtual hand while holding it, drag-and-drop can be performed, allowing movement across hand areas.

If the remote hand area and original hand area are the same sizes, the physical and virtual hands correspond to one-to-one mapping. When the size of the remote hand area is larger than the original hand area, the virtual hand motion can be expanded to allow the manipulation of large objects and large movements. On the other hand, when the size of the remote hand area is smaller, the virtual hand can perform more precise movements. The hand avatars in the area can be unimanual or bimanual depending on the situation. With two hands, the user can stretch objects, apply tools, or perform even more complex manipulations with full 10 fingers.

Hitchhiking provides fast and accurate operations by focusing on discrete interactions inside regions, as opposed to techniques like raycasting or HOMER which aim for uniform and global interactions. All existing hand-tracking applications, including tools, fluids, clay, etc., can be used without modification. It can also co-exist with other techniques such as raycasting and HOMER (one hand emits rays, another is directly manipulated, etc.) Furthermore, it alleviates the hand fatigue problem; wherever the virtual object is, the user can keep the physical hand in a comfortable position and use a displaced virtual hand instead.

On the other hand, this technique requires pre-positioning of hand areas, which might limit applicability. We assume application developers place hand areas in advance to where it might be helpful.

3 PRELIMINARY USER STUDY

The operability of HOMER and Hitchhiking were compared through a pre-user study conducted on five participants. The task was to grab an object, move it to another location, and place it in the correct position and orientation. All participants indicated that Hitchhiking was easier to operate than HOMER, and three of them indicated that it was less fatiguing than HOMER.

4 APPLICATIONS

Possible applications of this technique include the following tasks:

- (1) AR interaction in Daily life (Figure 1 left). Placement of AR objects and operation of IoT devices such as lightings.
- (2) XR windowing system (Figure 1 center). Placing one virtual hand on each window/application makes operating applications in various positions possible while keeping the hands comfortable, like on a 2D desktop.
- (3) Remote object manipulation in VR. By switching to larger hands, you can directly grasp and move buildings and terrain. Suitable for designing with a bird’s eye view.
- (4) Precise object manipulation in VR. By switching to smaller hands, precise operation of smaller objects is possible.
- (5) Manipulation of robot arms (Figure 1 right). This application is intended to be extension of Hitchhiking to the physical world and can be helpful in the control of prosthetic arms, or industrial use cases such as controlling cameras.

Application 1, 2, 3, and 5 can be experienced at the demo booth.

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REFERENCES

- Doug A. Bowman and Larry F. Hodges. 1997. An Evaluation of Techniques for Grabbing and Manipulating Remote Objects in Immersive Virtual Environments. In *Proceedings of the 1997 Symposium on Interactive 3D Graphics* (Providence, Rhode Island, USA) (I3D '97). Association for Computing Machinery, New York, NY, USA, 35–ff. <https://doi.org/10.1145/253284.253301>
- Mark R. Mine, Frederick P. Brooks, and Carlo H. Sequin. 1997. Moving Objects in Space: Exploiting Proprioception in Virtual-Environment Interaction. In *Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '97)*. ACM Press/Addison-Wesley Publishing Co., USA, 19–26. <https://doi.org/10.1145/258734.258747>
- Ken Pfeuffer, Benedikt Mayer, Diako Mardanbegi, and Hans Gellersen. 2017. Gaze + Pinch Interaction in Virtual Reality. In *Proceedings of the 5th Symposium on Spatial User Interaction* (Brighton, United Kingdom) (SUI '17). Association for Computing Machinery, New York, NY, USA, 99–108. <https://doi.org/10.1145/3131277.3132180>
- Ivan Poupyrev, Mark Billinghurst, Suzanne Weghorst, and Tadao Ichikawa. 1996. The Go-Go Interaction Technique: Non-Linear Mapping for Direct Manipulation in VR. In *Proceedings of the 9th Annual ACM Symposium on User Interface Software and Technology* (Seattle, Washington, USA) (UIST '96). Association for Computing Machinery, New York, NY, USA, 79–80. <https://doi.org/10.1145/237091.237102>
- Difeng Yu, Xueshi Lu, Rongkai Shi, Hai-Ning Liang, Tilman Dingler, Eduardo Velloso, and Jorge Goncalves. 2021. Gaze-Supported 3D Object Manipulation in Virtual Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 734, 13 pages. <https://doi.org/10.1145/3411764.3445343>